Lecture 9

SCALING LAWS - BASICS

CS336

Taking scaling seriously

Imagine the following scenario..

Your friend has given you ten thousand B200s for a month,
and asked you to build a good open source LM.

What do you do?

- Put together your infra team and distributed training framework (A2)
- Put together a great pretraining dataset (A4)
- Run a big model (but which one??) <- we are here.

Scaling isn’t easy

Wide or deep? How many heads? Which nonlinearity?

超参数太多选择了，我们需要选定真正能scaling up的

Name	Has pa...	Link	Year	Tokenizer type	Vocab count	Norm	Parallel Layer	Pre-norm	Position embedding	Activations	MoE	MLP factor	num_layers	model_dim
Original transformer	Yes	arxiv.org/abs....03762	2017	BPE	37000	LayerNorm	Serial	☐	Sine	ReLU	☐	4	6	
GPT	Yes	cdn.openai.com/res...er.pdf	2018	BPE	40257	LayerNorm	Serial	☐	Absolute	GeLU	☐	4	12	
GPT2	Yes	cdn.openai.com/bet...rs.pdf	2019	BPE	50257	LayerNorm	Serial	☑	Sine	GeLU	☐	4	48	
T5 (11B)	Yes	arxiv.org/abs....10683	2019	SentencePiece	32128	RMSNorm	Serial	☑	Relative	ReLU	☐	64	24	
GPT3 (175B)	Yes	arxiv.org/abs....14165	2020	BPE	50257	LayerNorm	Serial	☑	Sine	GeLU	☐	4	96	
mT5	Yes	arxiv.org/abs....11934	2020	SentencePiece	250000	RMSNorm	Serial	☑	Relative	GeGLU	☐	2.5	24	
T5 (XXL 11B) v1.1	Kind of	github.com/goo...d#t511	2020	SentencePiece	32128	RMSNorm	Serial	☑	Relative	GeGLU	☐	2.5	24	
Gopher (280B)	Yes	arxiv.org/abs....11446	2021	SentencePiece	32000	RMSNorm	Serial	☑	Relative	ReLU	☐	4	80	
Anthropic LM (not claude)	Yes	arxiv.org/abs....00861	2021	BPE	65536			☑			☐	4	64	
LaMDA	Yes	arxiv.org/abs....08239	2021	BPE	32000			☑	Relative	GeGLU	☐	8	64	
GPTJ	Kind of	huggingface.co/Ele...t-j-6b	2021	BPE	50257	LayerNorm	Parallel	☑	RoPE	GeLU	☐		28	
Chinchilla	Yes	arxiv.org/abs....15556	2022	SentencePiece	32000	RMSNorm	Serial	☑	Relative	ReLU	☐	4	80	
PaLM (540B)	Yes	arxiv.org/abs....02311	2022	SentencePiece	256000	RMSNorm	Parallel	☑	RoPE	SwiGLU	☐	4	118	
OPT (175B)	Yes	arxiv.org/abs....01068	2022	BPE	50272	LayerNorm	Serial	☑	Absolute	ReLU	☐	4	96	
BLOOM (175B)	Yes	arxiv.org/abs....05100	2022	BPE	250680	LayerNorm	Serial	☑	AliBi	GeLU	☐	4	70	
GPT-NeoX	Yes	arxiv.org/pdf...45.pdf	2022	BPE	50257	LayerNorm	Parallel	☑	RoPE	GeLU	☐	4	44	
GPT4 OPEN	Ad	arxiv.org/abs....08774	2023	BPE	100000			☐			☐			
LLaMA (65B)	Yes	arxiv.org/abs....13971	2023	BPE	32000	RMSNorm	Serial	☑	RoPE	SwiGLU	☐	2.6875	80	
LLaMA2 (70B)	Yes	arxiv.org/abs....09288	2023	BPE	32000	RMSNorm	Serial	☑	RoPE	SwiGLU	☐	3.5	80	
Mistral (7B)	Yes	arxiv.org/abs....06825	2023	BPE	32000	RMSNorm	Serial	☑	RoPE	SwiGLU	☐	3.5	32	

We could cargo cult things from existing LMs… but how do these get optimized in the first place?

Today: simple, predictive ‘laws’ for behaviors of LMs

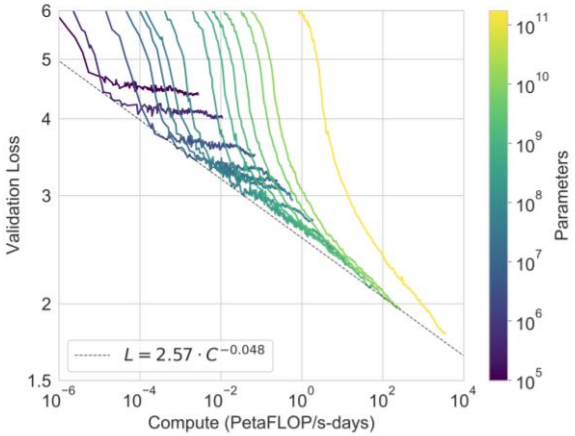
The approach -

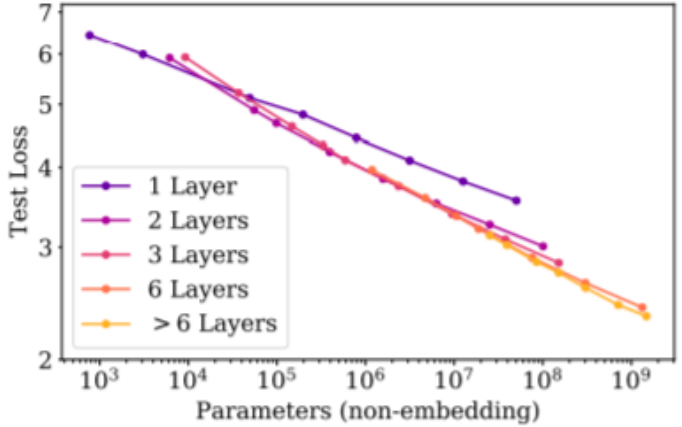
scaling laws which are simple, predictive rules for model performance

scaling laws are simple, predictive rules that allow us to go from great performance of small models to larger models.

Old and unpleasant: tune hyperparameters on big models
New (over?) optimism: tune on small models, extrapolate to large ones

先在小模型上面调超参，然后再将最好的超参组合扩充到大模型上





Part 1. Scaling laws, history and background.

- Data scaling as empirical sample complexities

- Initial forays into understanding neural scaling with data

Sample complexity and rates

Theorists have thought about ‘scaling’ for a long time..

$$\epsilon(\hat{h}) \leq \left(\min_{h \in H} \epsilon(h)\right) + 2\sqrt{\frac{1}{m} \log \frac{2k}{\delta}}$$

(learning in a finite set of k hypotheses)

Under the assumptions of Theorem 1.5, the rate of convergence of the estimator $\hat{p}_n(x_0)$ is $\psi_n = n^{-\frac{\beta}{2\beta+1}}$, which means that for a finite constant C and for all $n \geq 1$ we have

$$\sup_{p \in \mathcal{P}(\beta, L)} \mathbb{E}_p \left[\left(\hat{p}_n(x_0) - p(x_0) \right)^2 \right] \leq C \psi_n^2.$$

(generative modeling for smooth densities)

But these are upper bounds, not actual, realized loss values.

给定参数量，我们可以知道一个能收敛的estimator的loss的上界。
这是有理论支撑的。

https://www.cs.cmu.edu/~epxing/Class/10701/slides/lecture16-VC.pdf
Hall, 1989

Earliest (data) scaling law paper – 1993

Learning Curves: Asymptotic Values and Rate of Convergence

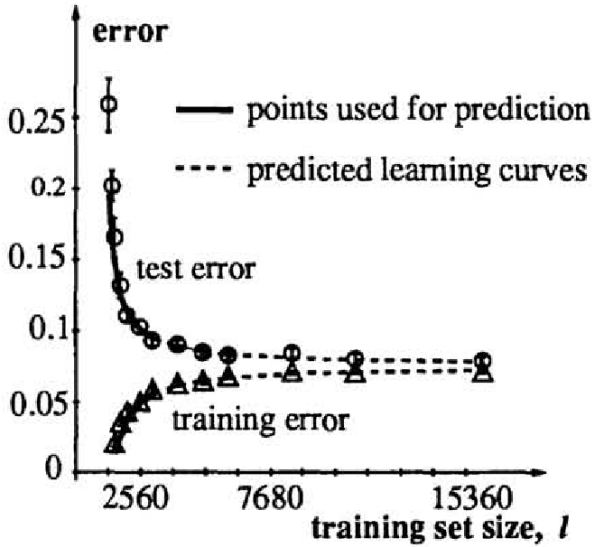
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AT&T Bell Laboratories
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Abstract

Training classifiers on large databases is computationally demanding. It is desirable to develop efficient procedures for a reliable prediction of a classifier's suitability for implementing a given task, so that resources can be assigned to the most promising candidates or freed for exploring new classifier candidates. We propose such a practical and principled predictive method. Practical because it avoids the costly procedure of training poor classifiers on the whole training set, and principled because of its theoretical foundation. The effectiveness of the proposed procedure is demonstrated for both single- and multi-layer networks.

A typical example of learning curves is shown in Fig. 2. The test error is always larger than the training error, but asymptotically they reach a common value, a. We model the errors for large sizes of the training set as power-law decays to the

$$\mathcal{E}_{\text{test}} = a + \frac{b}{l^{\alpha}} \qquad \text{and} \qquad \mathcal{E}_{\text{train}} = a - \frac{c}{l^{\beta}}$$



拟合：训练集大小和损失之间的关系

Early history of scaling laws – data scaling

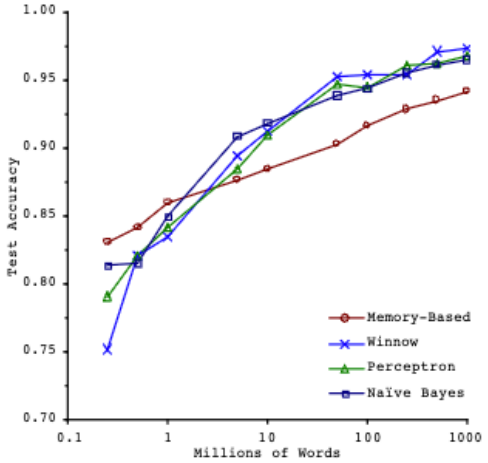


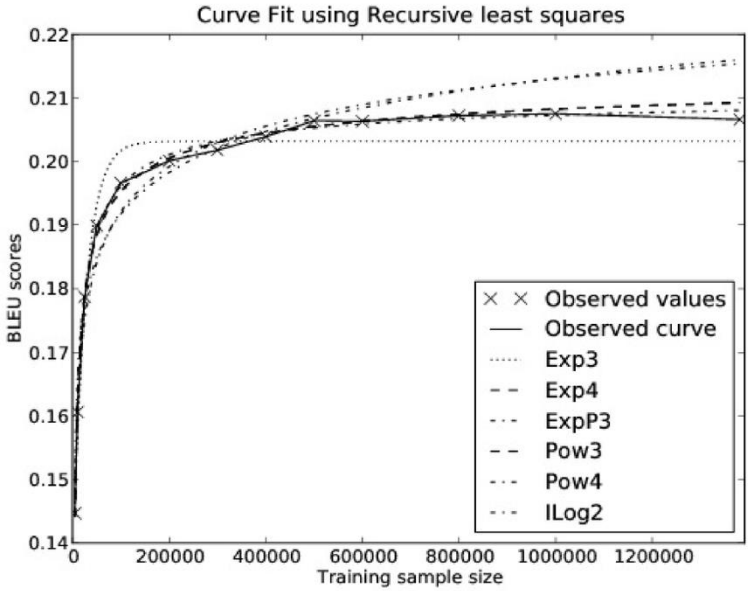
Figure 1. Learning Curves for Confusion Set Disambiguation

these results suggest that we may want to reconsider the trade-off between spending time and money on algorithm development versus spending it on corpus development. At least for the problem of confusable disambiguation, none of the learners tested is close to asymptoting in performance at the training corpus size commonly employed by the field.

Log-linear scaling with data [Banko and Brill ‘01]

与其提升算法，收集更多数据是否更加有用且简单?

Early history of scaling laws – data scaling



Model	Formula
Exp_3	$y = c - e^{-ax+b}$
Exp_4	$y = c - e^{-ax^{\alpha}+b}$
ExpP_3	$y = c - e^{(x-b)^{\alpha}}$
Pow_3	$y = c - ax^{-\alpha}$
Pow_4	$y = c - (-ax+b)^{-\alpha}$
ILog_2	$y = c - (a/\log x)$

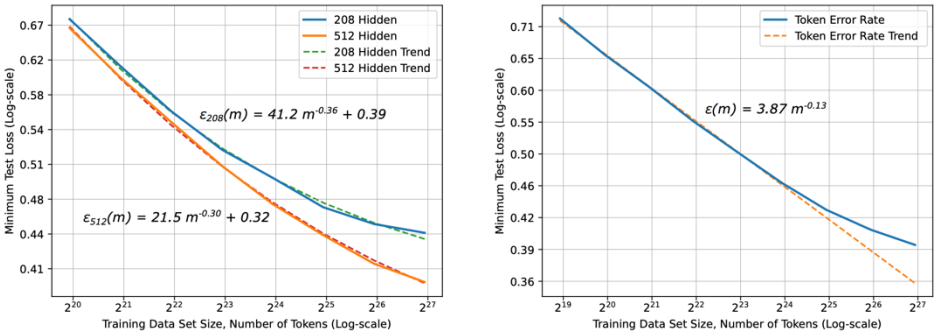
Table 1: Curve families.

基本上是Pow3和Pow4的scaling recipe

Early tests of functional forms

Kolachina et al 2012 – power law relation between data and downstream performance

Hestness et al 2017



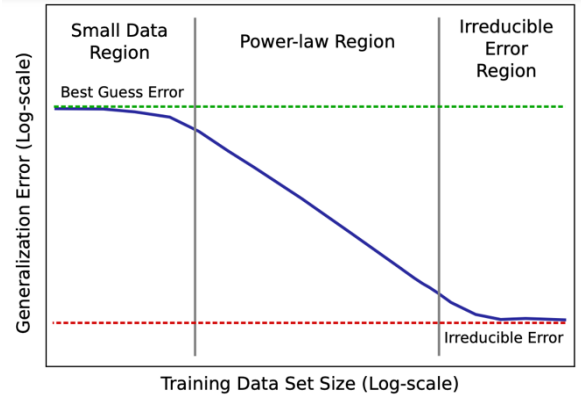


Figure 1: Neural machine translation learning curves. Left: the learning curves for separate models follow $\varepsilon(m) = \alpha m^{\beta_g} + \gamma$. Right: composite learning curve of best-fit model at each data set size.

Earliest 'large scale neural' scaling work: Hestness 2017

Predictable scaling on many tasks (MT, LM, Speech) and hypothesized scaling shape.

Hestness II

Very ahead of its time..

“Emergence”
智能的涌现

> Although small data set testing may be possible, it can be difficult to ensure that training data is large enough to see the power-law learning curve region. We have found that models with poor optimizer parameterization or model priors/initialization show accuracy cliffs, where accuracy is only as good as best guessing, but the model trains on enough data to be in the power-law region. Researchers must take great care when defining a "large enough" training set for small data testing. We leave the methodology for defining such a training set to future work.

Scaling by compute

> **Computational Limits:** If we have identified a desirable model to scale to larger training sets, the next potential limitation is the speed of computation. In some cases, training large models on very large data sets would take months or years of critical path compute time, making these training runs impractical for any real world problem on existing systems. However, predictable learning and model size curves may offer a way to project the compute requirements to reach a particular accuracy level. The compute requirements could inform decisions about how to scale computational capacity to unlock these compute-limited applications.

Speed = accuracy

> **The Performance-Accuracy Trade-off:** Many DL software and hardware techniques impose a trade-off between model accuracy and the speed of computation. Learning curves and model size growth can indicate whether these techniques could regain lost accuracy by improving the speed of computation. For example, low-precision computation/quantization and sparse models give up some model accuracy (e.g., up to 20%) in order to improve compute throughput. If the compute throughput improvements allow DL developers to train larger models on larger data sets, these accuracy losses might be easily recoverable.

Part 2. Neural (LLM) scaling behaviors

1.Data vs performance

“Are there simple rules that determine how data affects performance?”

2. Data vs model size

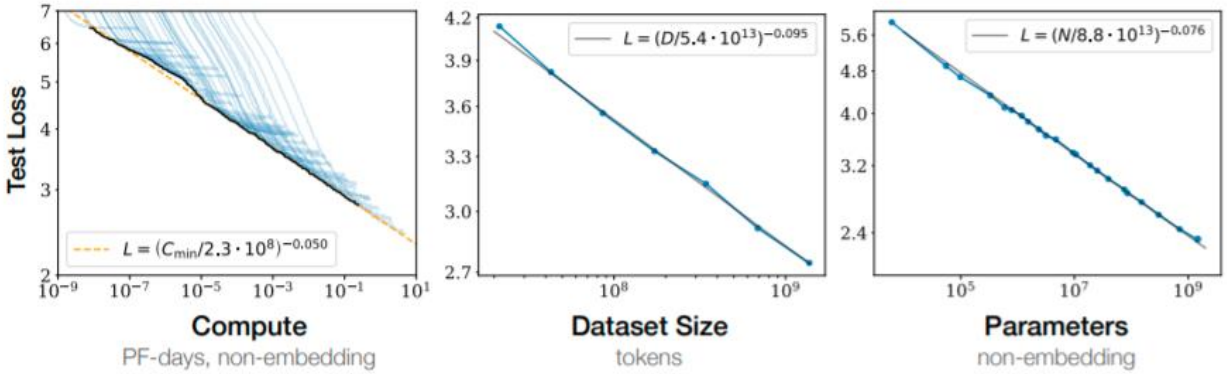
Do we train on more data or bigger models?

3. Hyper-parameters vs performance

“How should we set hyperparameters on the big model??”

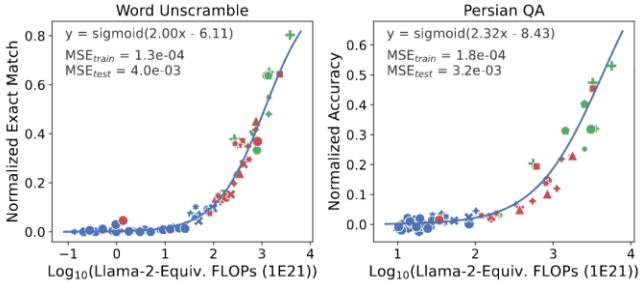
Scaling laws – power law relationships for many factors

These scaling laws hold on *many* different kind of phenomena!



这张图的横坐标和
纵左表，都是Log形式

They even hold in non standard settings





[Kaplan+ 2020]

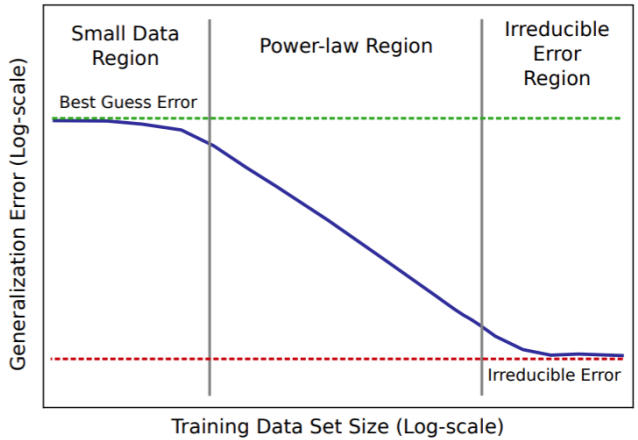
Data vs performance

What's a data scaling law?

Data scaling laws : simple formula that maps dataset size (n) to error

What do we expect out of scaling laws?

Monotonic, logistic-like curves

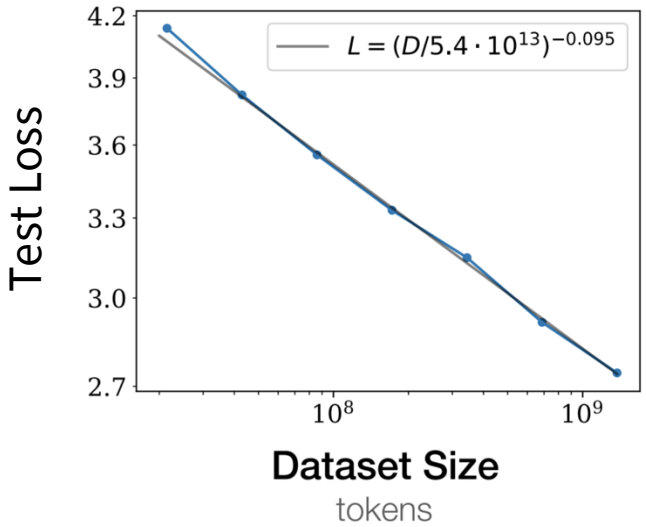


#TODO：这张图具体的含义详细解释下。以及这里的irreducible error指的是什么？ [Hestness+ 2017]

Data scaling laws for language models

First, an empirical observation

Loss and dataset size is linear on a log-log plot



“Scale-free” or
“Power law”

横纵轴都是log形式

(For language modeling, from Kaplan+ 2020)

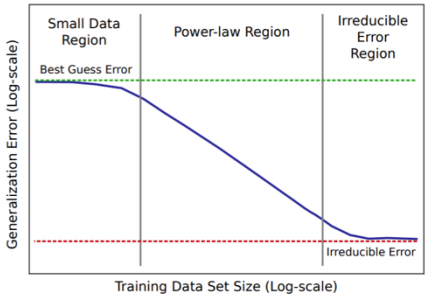
Conceptual foundations of data scaling laws.

Q: Why do scaling laws show up?

We know error should be monotone →

#TODO：这里是啥意思？monotone是什么意思？为啥error should be monotone?

But why is it a power law / linear in log-log?



A (?): Estimation error naturally decays polynomially.

But this answer may take a moment to understand. Let's work through an example.

Example: If our task is to estimate the mean of a dataset, what's the scaling law?

Toy example: mean estimation

Input: $x_1 \ \ldots x_n \ \sim N(\mu, \sigma^2)$

Task: estimate the average as $\hat{\mu} = \frac{\sum_i x_i}{n}$

What's the error? By standard arguments..

$$\mathrm{E}[(\hat{\mu} - \mu)^2] = \frac{\sigma^2}{n}$$

这也是scaling law

This is a 'scaling law'

$$\log(Error) = -\log n + 2\log \sigma$$

这里的log scaling law的系数是-1

More generally, any polynomial rate $1/n^{\alpha}$ is a scaling law

#TODO polunomial rate指的是啥? 这里是啥意思?

Scaling law exponents: an intriguing mystery

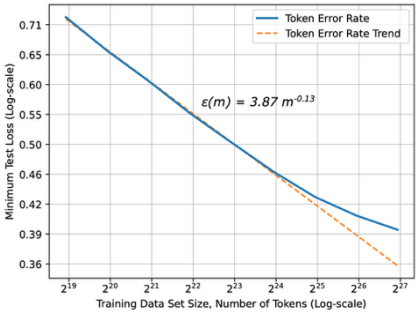
Fact: Similar arguments show most 'classical' models (regression, etc) have $\frac{1}{n}$ scaling

传统的模型：对于1/n scaling，那么y 正比于 1/n, 则logy正比于-logn，即系数是-1。下式的x就是n的意思。

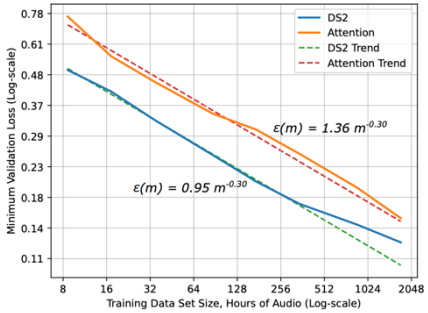
This means we should see $y = -x + C$

What do we find in neural scaling laws?

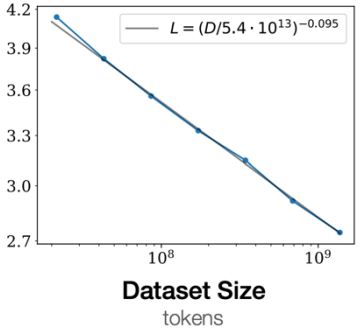
但是实际上不是-1，而是-0.13，-0.30这类数字，说明loss相比1/n scaling，下降得慢很多



Machine translation



Speech



Language modeling

Very different from predictions.. Why might this be?

Detour: scaling laws for (nonparametric) learning

Neural nets can approximate arbitrary functions. Lets turn that into an example.

Input: $x_1 \ldots x_n$ uniform in 2D unit box. $y_i = f(x_i) + N(0,1)$
Task: estimate f(x)

Approach: cut up the 2D space into boxes with length $n^{-\frac{1}{4}}$

What's our estimation error?

#TODO：和前面得1/n scaling 的情况，这里是什么不同导致其他系数的情况?

Informally, we have $\sqrt{n}$ boxes, each box gets $\sqrt{n}$ samples.

$$Error \approx \frac{1}{\sqrt{n}} + (other\ smoothness\ terms)$$

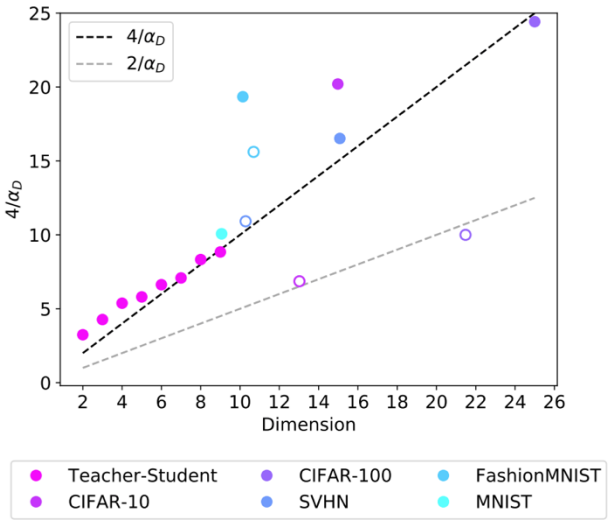
In d-dimensions, this becomes $Error = n^{-1/d}$ - **This means scaling is** $\boldsymbol{y = -\frac{1}{d}x + C}$

Takeaway: flexible 'nonparametric' learning has dimension dependent scaling laws.

Intrinsic dimensionality theory of data scaling laws

Some have made the following argument (Bahri 2021)

1. Scaling laws arise due to polynomial rates of learning $\frac{1}{n^{\alpha}}$
2. Some argue the slope α is closely connected to the *intrinsic dimensionality* of the data.



But estimators of intrinsic dimension are sketchy, and this is not airtight..

Other data scaling laws

Data scaling thus far: how does dataset size relate to performance?
Related question: how does dataset *composition* affect performance

- Picking optimal data mixture using small scale models
- Deciding whether to repeat data or not
- Combing the two and balancing quality with repetition rate

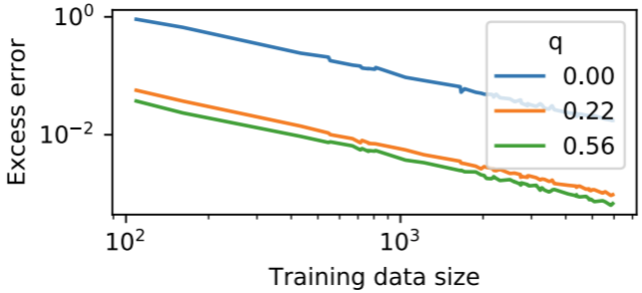
Other advanced data scaling law: distribution shift

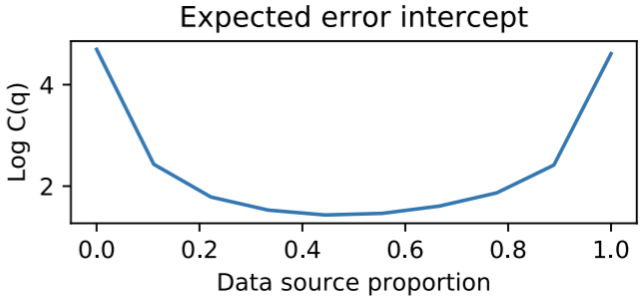
Data scaling thus far: how does dataset size relate to performance?

Related question: how does dataset *composition* affect performance

A: Data composition affects the offset, not the slope.

These 'distribution shift' scaling laws can tell us about the importance of collecting diverse data!





q可以理解为数据中，一种类型的数据占比，表示数据组成。

数据的组成影响的是截距项，而不是斜率。

[Hashimoto 2021]

In practice: data mixture selection via scaling is hard

Data Mixing Laws: Optimizing Data Mixtures by Predicting Language Modeling Performance

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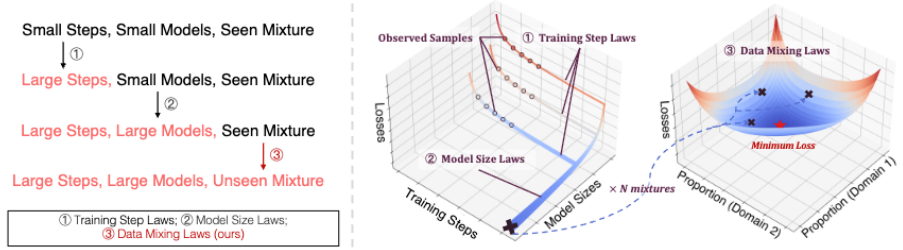


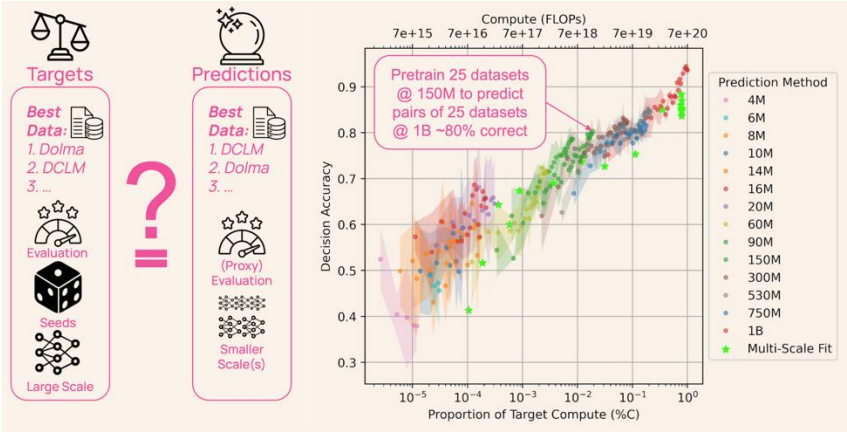
Figure 1: Illustration on our pipeline to optimize data mixture. **Left:** Our pipeline takes three steps. Starting from small-scale training results, the three steps use the scaling laws of training steps, model sizes, and data mixing laws to predict model performance on large steps, large models, and unseen mixtures, respectively. **Right:** Visualization of the three-step pipeline to predict model performance on the target model size, training step, and mixtures.

Natural idea – build data scaling laws

DataDecide

How to Predict Best Pretraining Data with Small Experiments

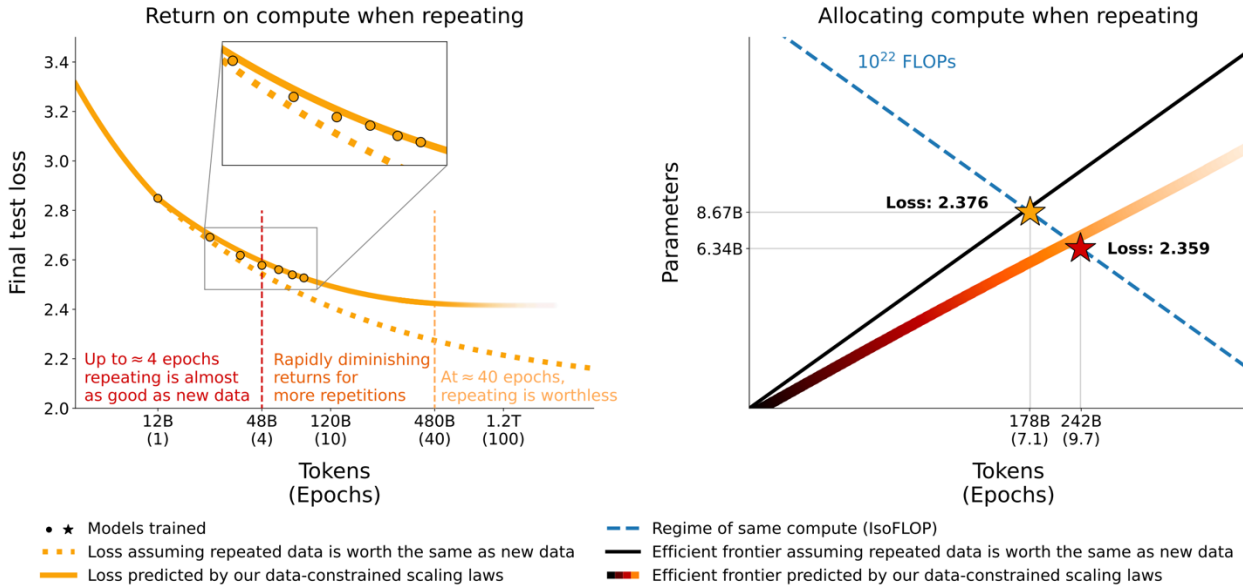
Ian Magnusson[*12] Nguyen Tai[*3] Ben Bogin[*1] David Heineman[1] Jena Hwang[1] Luca Soldaini[1] Akshita Bhagia[1]
Jiacheng Liu[12] Dirk Groeneveld[1] Oyvind Tafjord[1] Noah A. Smith[12] Pang Wei Koh[12] Jesse Dodge[1]



Empirical eval – just take the best small dataset

Scaling laws under data repetition

In practice, we have finite data – how does repeating examples affect scaling?



Scaling Data-Constrained Language Models

Niklas Muennighoff [1] Alexander M. Rush [1] Boaz Barak [2] Teven Le Scao [1]
Aleksandra Piktus [1] Nouamane Tazi [1] Sampo Pyysalo [3] Thomas Wolf [1] Colin Raffel [1]
[1] Hugging Face [2] Harvard University [3] University of Turku
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$$D' = U_D + U_D R_D^* (1 - e^{\frac{-R_D}{R_D^*}}) .$$

D' = Effective data
Ud = Unique tokens
Rd* = Constant
Rd = Repetition

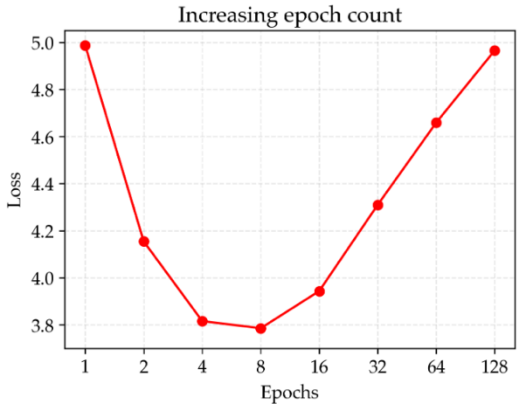
Scaling laws in *compute unbounded* settings

Pre-training under infinite compute

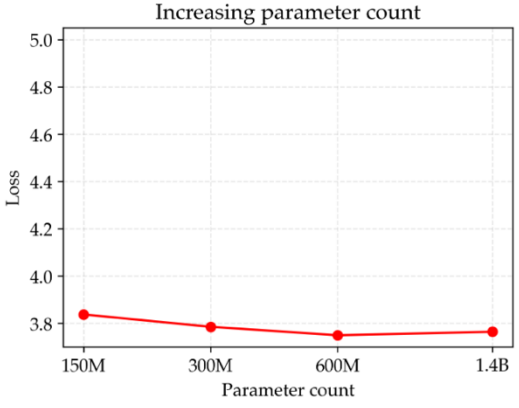
Konwoo Kim∞, Suhas Kotha∞, Percy Liang, Tatsunori Hashimoto
Stanford University

Important notes:

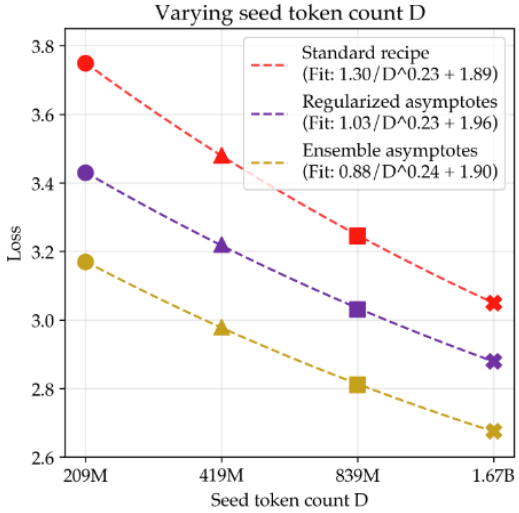
1. Scaling laws can ‘break’ if you blindly apply it
2. Scaling laws are *lower bounds* so you can always potentially do better



Tuned H	1	8	128
Learning rate	1e-3	1e-3	3e-3



Tuned H	150M	300M	600M	1.4B
Learning rate	3e-3	1e-3	1e-3	3e-4
Epoch count	8	8	4	4



几种不同的模型，包括集成模型，slope都差不多

#TODO：这里slope差不多说明了什么？有什么启发和作用?

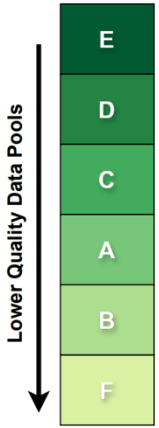
Data selection scaling and accounting for finiteness

Given that repeated data is less valuable..

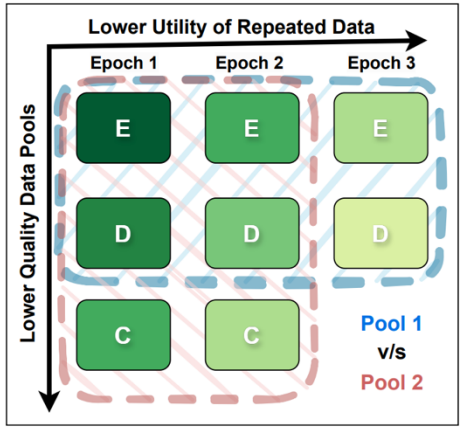
Data selection should then be adaptive to scale !

Scaling Laws for Data Filtering—
Data Curation *cannot* be Compute Agnostic

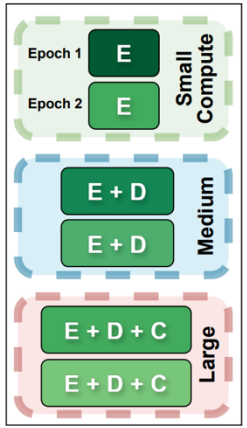
Sachin Goyal*† Pratyush Maini*†
Zachary C. Lipton† Aditi Raghunathan† J. Zico Kolter†‡
Carnegie Mellon University† Bosch Center for AI‡
{sachingoyal,pratyushmaini,zlipton,raditi,zkolter}@cmu.edu



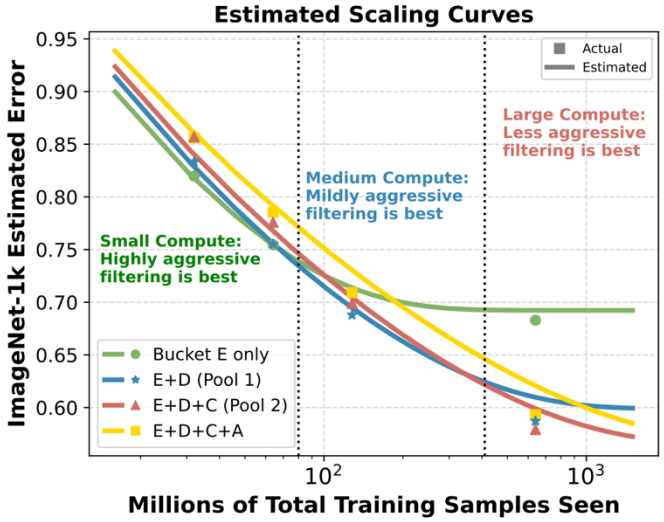
Web Data is Non-Homogenous



Quality-Quantity Tradeoff (QQT) for Data Filtering



Best Data Pool Changes with Total Compute



拥有越大的flops预算，我们就会更加倾向于加入更多的low quality的data
所以最佳的数据选取，需要和flops预算挂钩

Recap: data scaling laws

- Remarkably linear relationship between log-data size and log-error
- Holds across domains and models
- Theory understanding: similar to generalization bounds: mean estimation example
- Applications: data collection / curation

Scaling laws for model engineering

Now for what I promised at the start: **model scaling!**

Our motivation: how can we efficiently design huge LMs?

- LSTMs vs Transformers
- Adam vs SGD

How should we allocate our limited resources?

- Train models longer vs train bigger models?
- Collect more data vs get more GPUs?

Scaling laws provide a simple procedure to answer these.

Hyperparameter questions

We'll consider some of these choices in the context of the classic Kaplan scaling paper

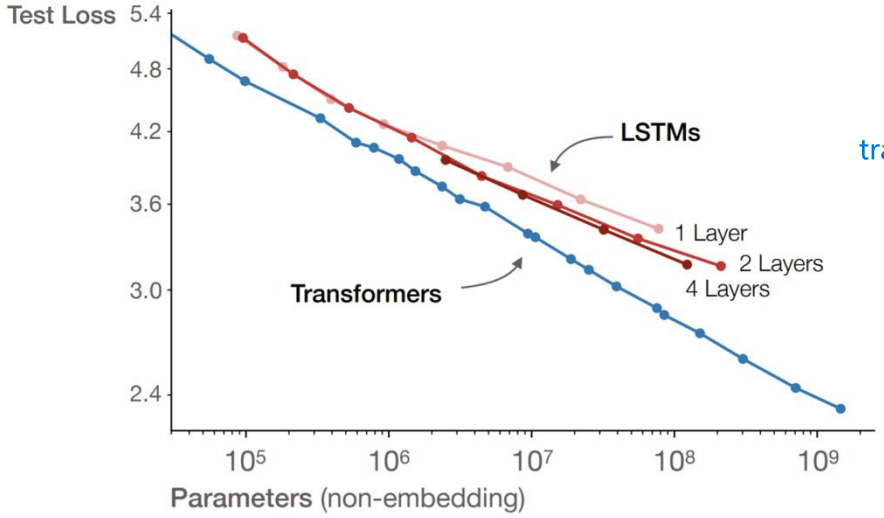
- Architecture
- Optimizer
- Aspect ratio / depth
- Batch size

1. Architecture: transformers vs LSTMs

Q: Are transformers better than LSTMs?

Brute force way: spend tens of millions to train a LSTM GPT-3

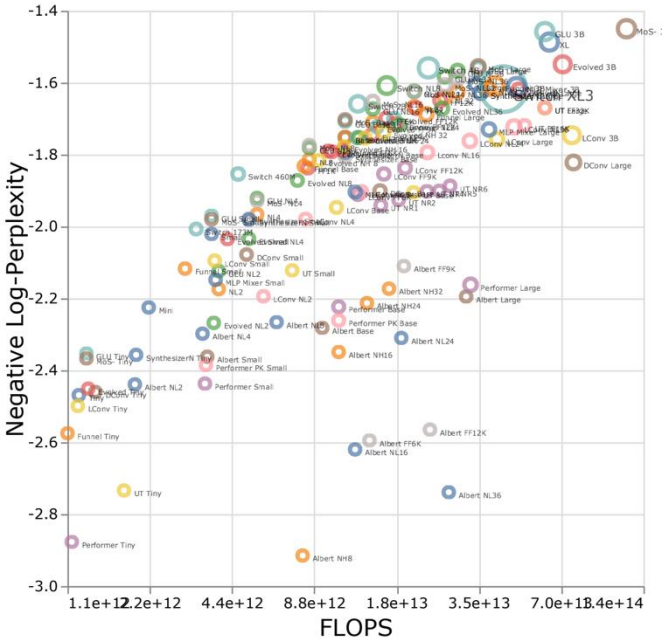
Scaling law way:

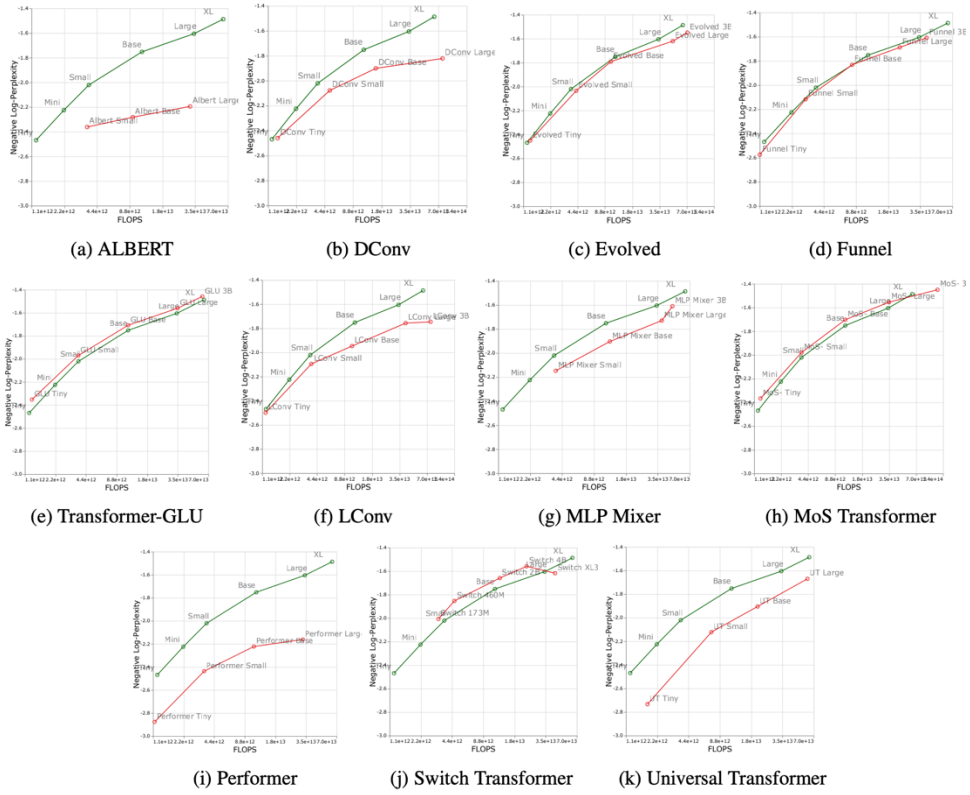


transformers相比LSTM，能scaling得更好

[Kaplan+ 2021]

1. Many architectures



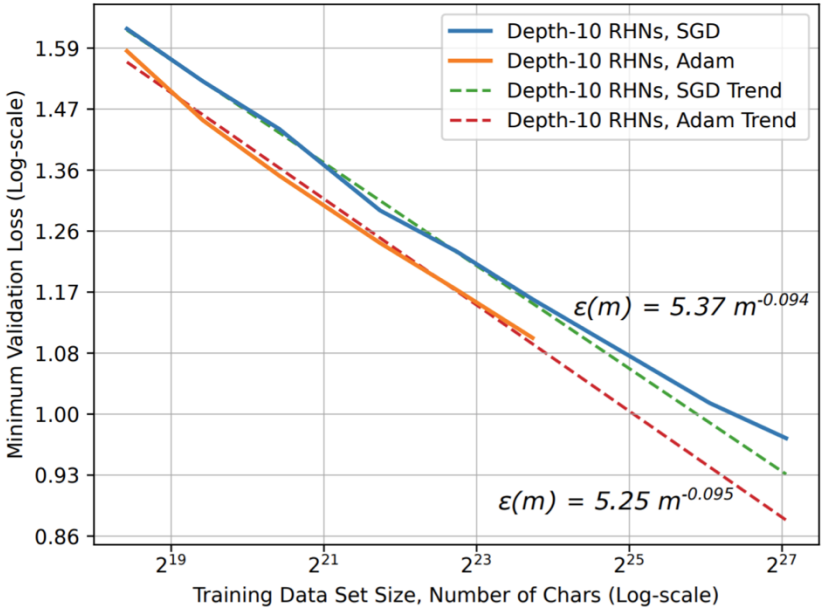


Cross-architecture scaling in Tay et al 2022.

很多之前的架构都有scaling的趋势。所以很多人都认为，一个架构如果无法看出scaling law，那么这个架构就不够好。

2. Optimizer choice

What about ADAM vs SGD?



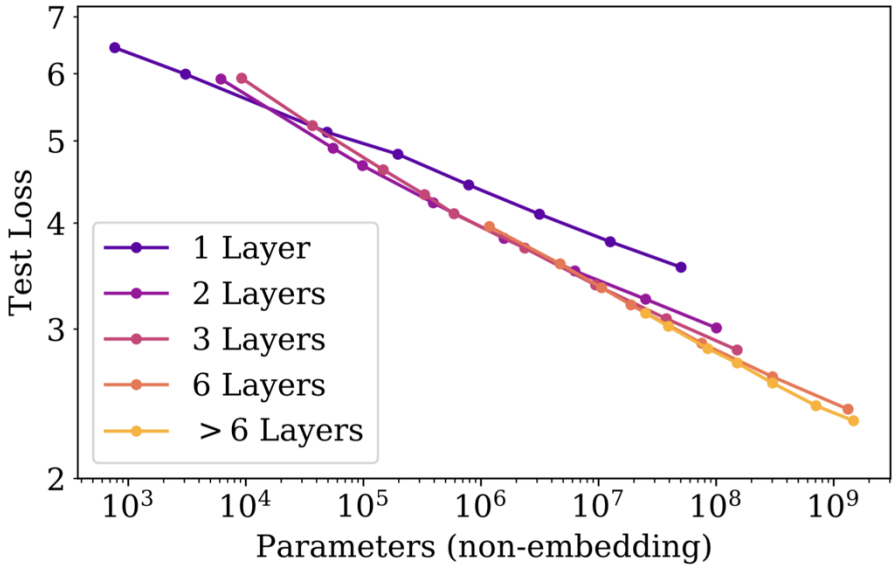
很多时候，无论做了多大的架构上的改进，斜率都差不多，即scaling trend没有什么改变。改进点在于截距项。

[Hestness+ 2017]

(Note, this is in 2017, so pre-transformers. RHN is recurrent highway nets)

3. Depth/Width: Number of layers

Does depth or width make a huge difference?



- 1 vs 2 layers makes a huge difference.
- More layers have diminishing returns below 10^7 params

3. Depth/Width: and other Transformer hypers

Do hyperparameters like the aspect ratio depend on scale?

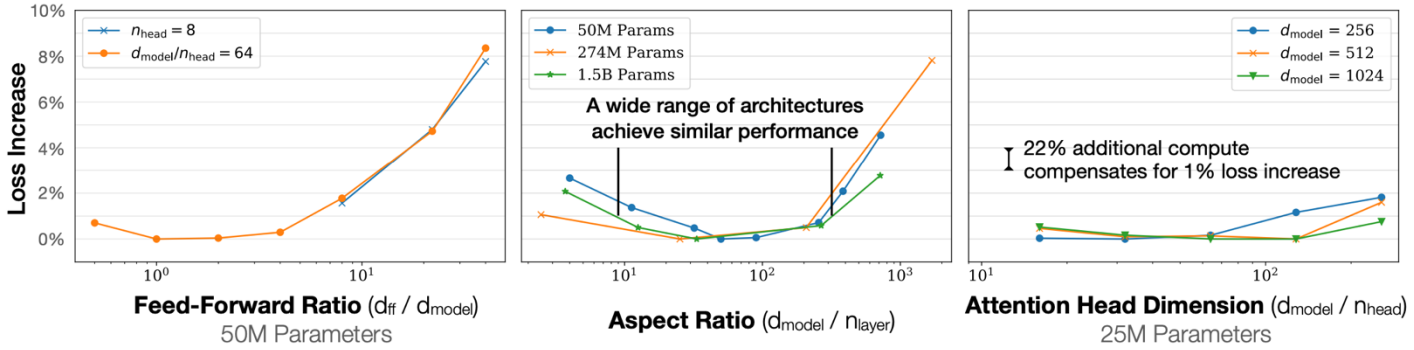
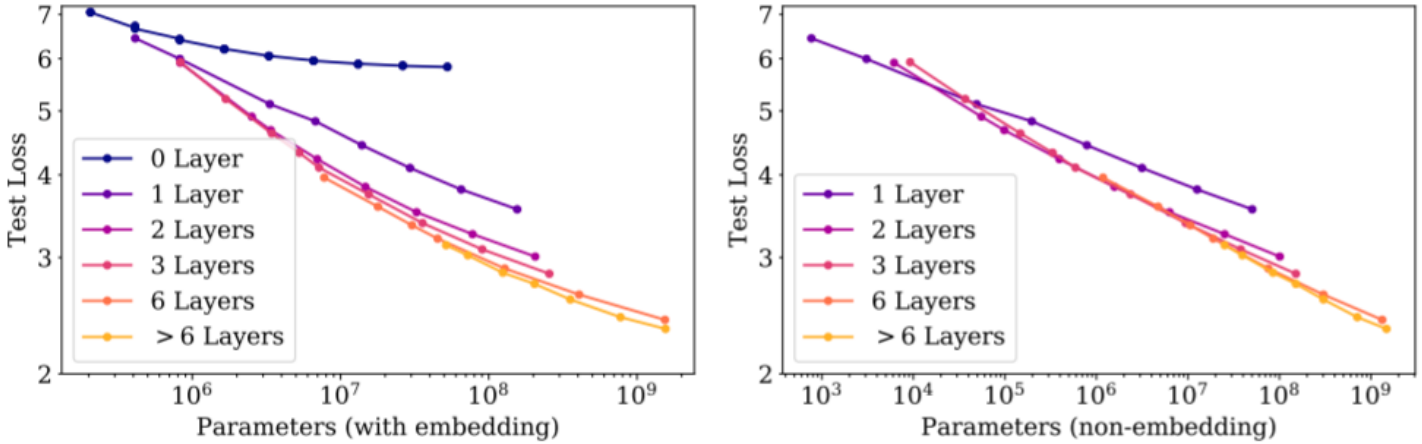


Figure 5 Performance depends very mildly on model shape when the total number of non-embedding parameters N is held fixed. The loss varies only a few percent over a wide range of shapes. Small differences in parameter counts are compensated for by using the fit to $L(N)$ as a baseline. Aspect ratio in particular can vary by a factor of 40 while only slightly impacting performance; an $(n_{\text{layer}}, d_{\text{model}}) = (6, 4288)$ reaches a loss within 3% of the $(48, 1600)$ model used in [RWC+19].

3. Depth/Width: But not all parameters are made equal

We've been thinking about 'parameters' but not all parameters are equal



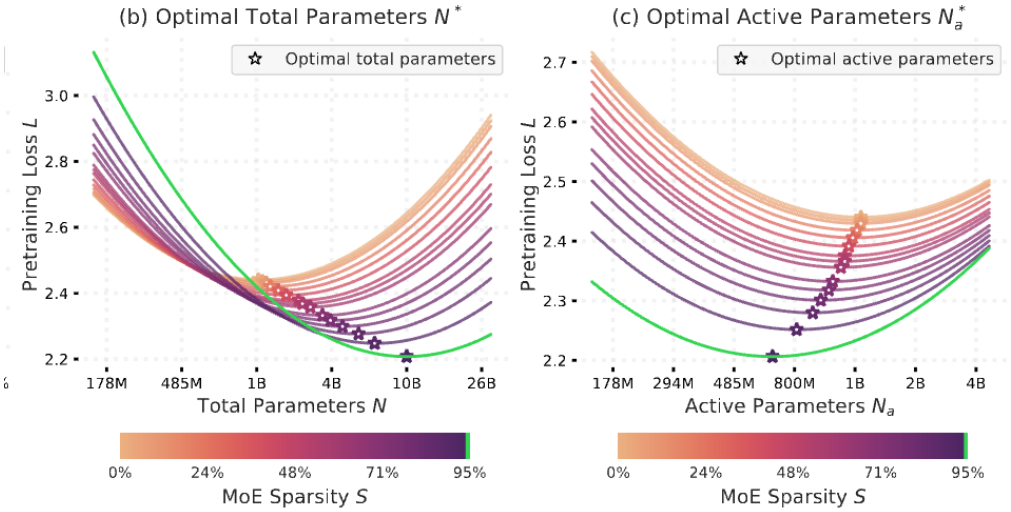
Embedding layer parameters don't behave the same!

所以重要的是，哪个方式才是可预测的、能够得到predictable scaling的方式

Related: recent papers on scaling laws for mixtures of experts.

Side note – ‘Value of parameters’

With MoEs, we expect parameter scaling to change

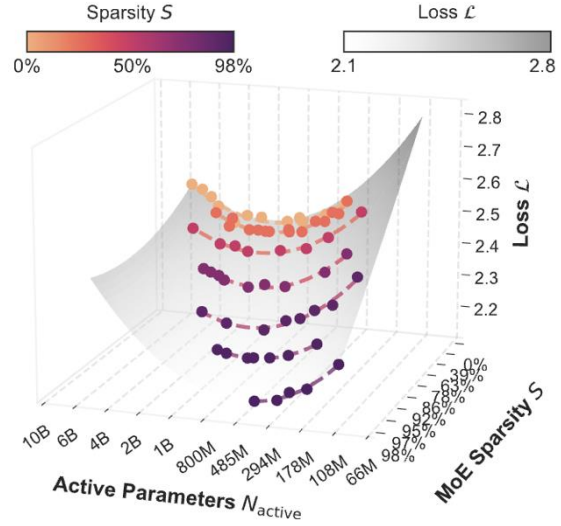


scaling是parameters和activeated parameters的函数

图中这几条线的flops总预算都是相同的。
则可以解答问题：对于MoE模型，给定flops总预算，我要将其花在总参数量上，还是花在激活的参数量上？控制总flops相同，我是否需要增加sparsity?

Parameters vs FLOPs: Scaling Laws for Optimal Sparsity for Mixture-of-Experts Language Models

Samira Abnar* Apple
Harshay Shah* MIT
Dan Busbridge Apple
Alaaeldin Mohamed Elnouby Ali Apple
Josh Susskind Apple
Vimal Thilak* Apple



(b) IsoFLOP surface over sparsity and active parameters

4. Batch size: Critical batch size

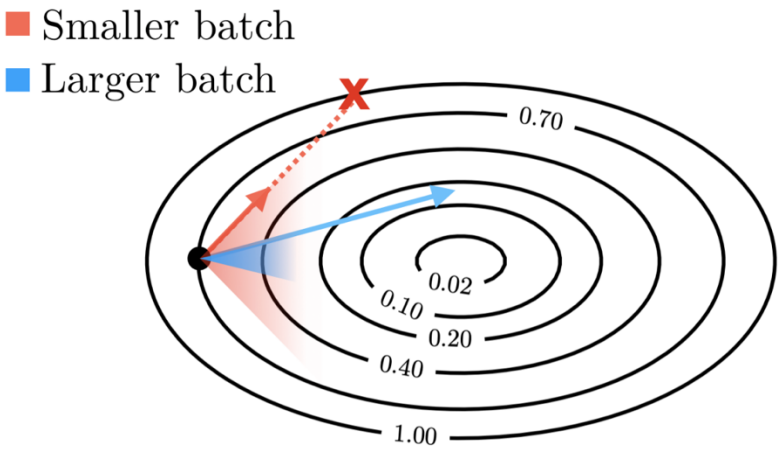
An Empirical Model of Large-Batch Training

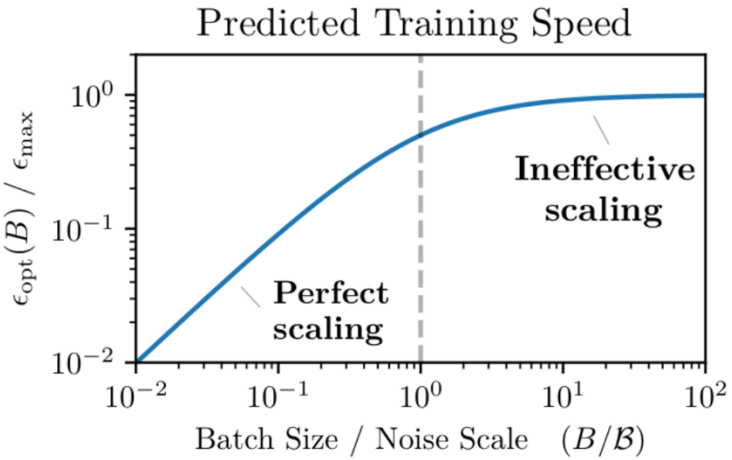
Sam McCandlish* OpenAI sam@openai.com

Jared Kaplan Johns Hopkins University, OpenAI jaredk@jhu.edu

Dario Amodei OpenAI damodei@openai.com

and the **OpenAI Dota Team**†





#TODO：这里的noise scale是啥意思？这张图的横坐标和纵坐标分别是啥?

Batch size – known to have strong diminishing returns past a certain point.

Critical batch = min number of examples before diminishing returns (what does that mean?)

这也是一个经验法则。

#TODO：解释下critical batch的含义，以及为啥会有diminishing returns?

4. Critical batch size definitions..

What is critical batch size, more precisely?

- Pick a target loss and train 1). Steps needed (S), 2). Examples needed.
- Sweep over batches

steps * batch_size = examples

The curve should follow roughly..

这个式子说的是S和E的trade-off。改变batch_size就会出现不同的steps和examples的组合

$$\frac{S}{S_{\min}} - 1 = \left(\frac{E}{E_{\min}} - 1\right)^{-1}$$

S：steps
E：Examples

Fit for S_{min}, E_{min} on this curve and pick

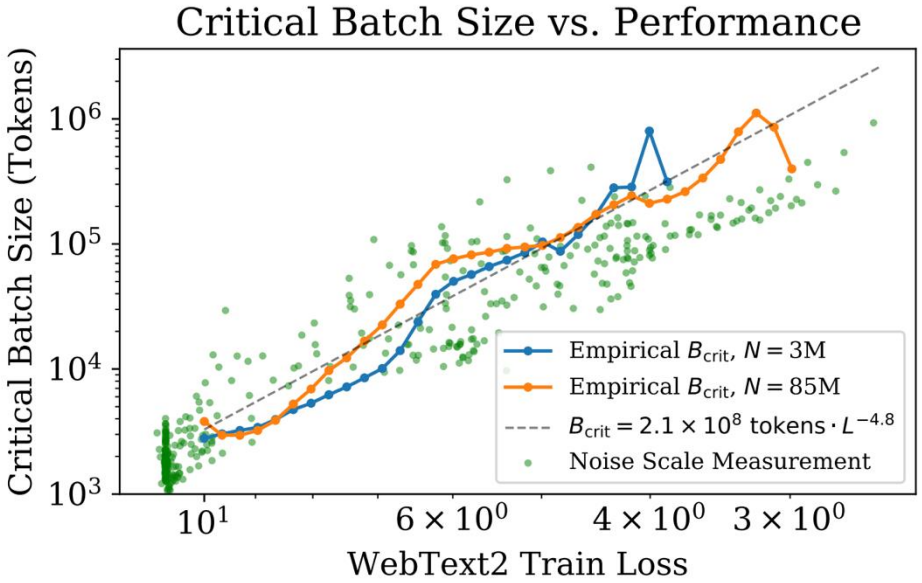
#TODO：详细解释下这个式子，包括每个符号的含义，和式子的含义和意义等。另外，Smin和Emin是如何计算出来的?

$$\mathcal{B}_{\text{crit}} = \frac{E_{\min}}{S_{\min}}.$$

This balances both sides of the equation, giving roughly 2x the steps / passes optimal

(this is claimed to be close to the ratio of the trace of the gradient covariance and squared norm of the gradient)

4. Batch size: critical batch size

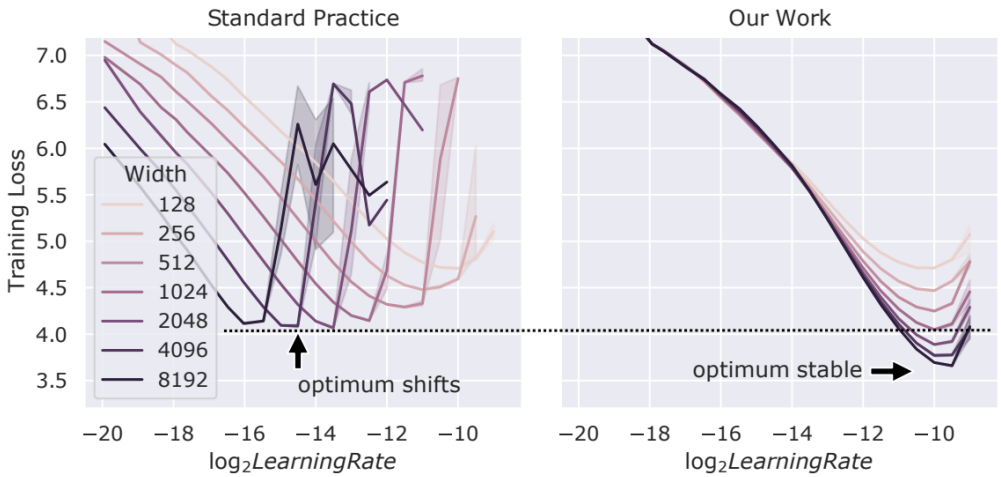


#TODO：详细解释下这张图，以及下方的公式。我没看懂含义和这张图想要说明什么。

The smaller the loss target,
The bigger the batch

$$C_{\min}(C) \equiv \frac{C}{1 + B/B_{\text{crit}}(L)} \qquad \text{(minimum compute, at } B \ll B_{\text{crit}}\text{)}$$

5. Learning rates: muP and scale-aware LR choices



Yang et al 2022

Table 2. μP function for a model M' that is r times the widths of M. If a parameter tensor has 2 dimensions that goes infinite when the model width goes infinite, it is "matrix-like" (*e.g.,* a fully-connected hidden layer); if the number is 1 or 0, it belongs to the "others" class. Note that embedding layers are "others". "Output" means the layer that maps an infinite dimension to a finite dimension, which is the word decoding layer (lm_head) in Transformers. A multiplier is a constant multiplied by a parameter tensor, which has a similar function to softmax temperature.

Hyperparameter (weight)	M	$M' \sim r$
AdamW learning rate (matrix-like)	l	l/r
AdamW learning rate (others)	l	l
Initialization variance (matrix-like)	σ	σ/r
Initialization variance (others)	σ	σ
Multiplier (output)	τ	τ/r
Multiplier (others)	τ	τ

Yao et al 2024

If we naively scale up – optimal learning rate depends on scale.
We need *scaling aware* initialization and learning rate scaling

#TODO：详细解释下两幅图的区别，以及右图的做法μP，到底是什么含义？

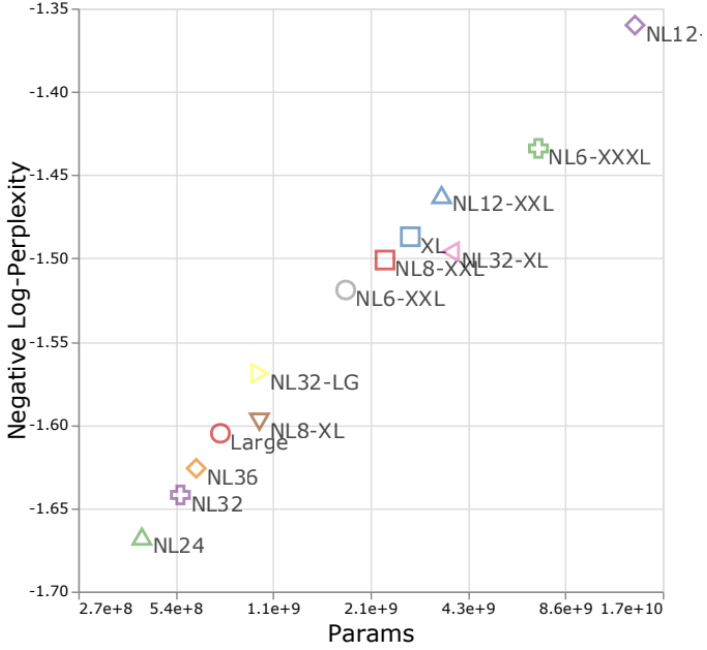
Caution – scaling behaviors can differ downstream

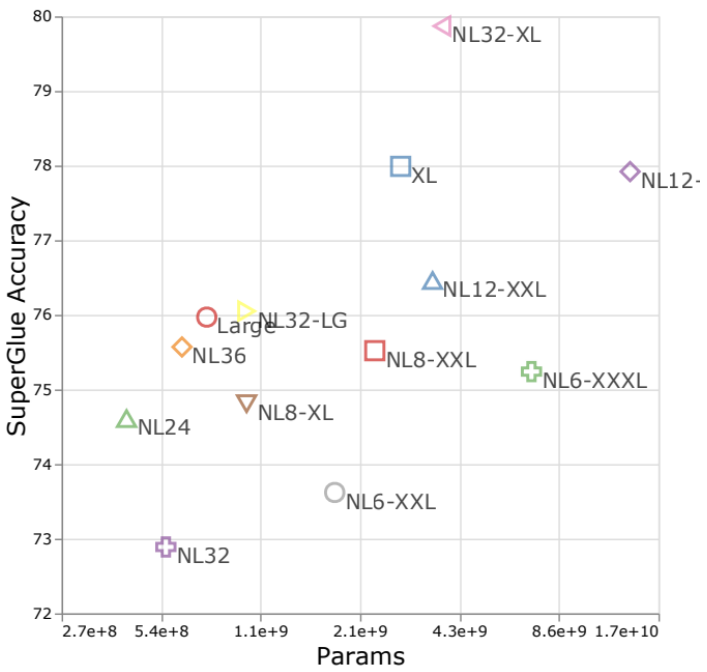
Thus far: scaling is predictable and depends mainly on parameters

Catch: downstream scaling can often be much less predictable

不同指标会到来不同的scaling trend。perplexity指标表现出的良好的scaling trend放在下游accuracy等指标上可能就不一致了

perplexity一般是pretrain阶段的指标，而accuracy一般是post-train阶段的指标。





Tay et al 2023

Some surprising takeaways

The effect of hyperparameters on big LMs can be predicted *before* training!

- Optimizer choice
- Model depth
- Architecture choice

The scaling law based design procedure.

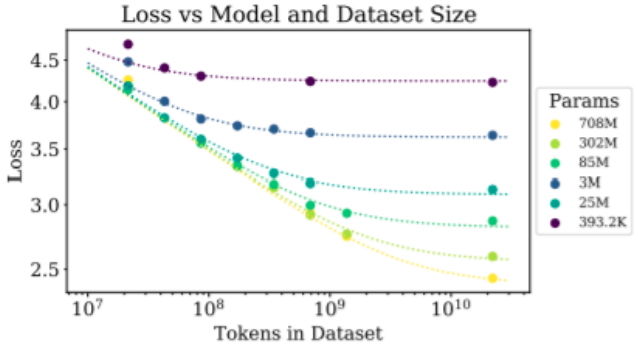
1. Train a few smaller models
2. Establish a scaling law (e.g. ADAM vs SGD scaling law)
3. Select optimal hyperparam based on the scaling law prediction.

这个步骤是一个通用的步骤，无论对于哪个超参数的调整，都可以用上。

One important use of scaling laws

Q: Do we need more data or bigger models?

Clearly, lots of data is wasted on small models



Joint data-model scaling laws describe how the two relate

From Rosenfeld+ 2020,

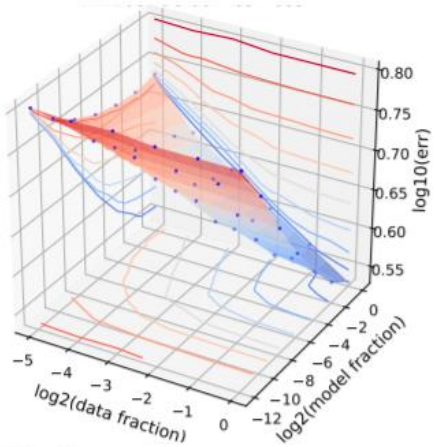
n和m一个是数据量，一个是模型参数

$$Error = n^{-\alpha} + m^{-\beta} + C$$

From Kaplan+ 2020

$$Error = [m^{-\alpha} + n^{-1}]^{\beta}$$

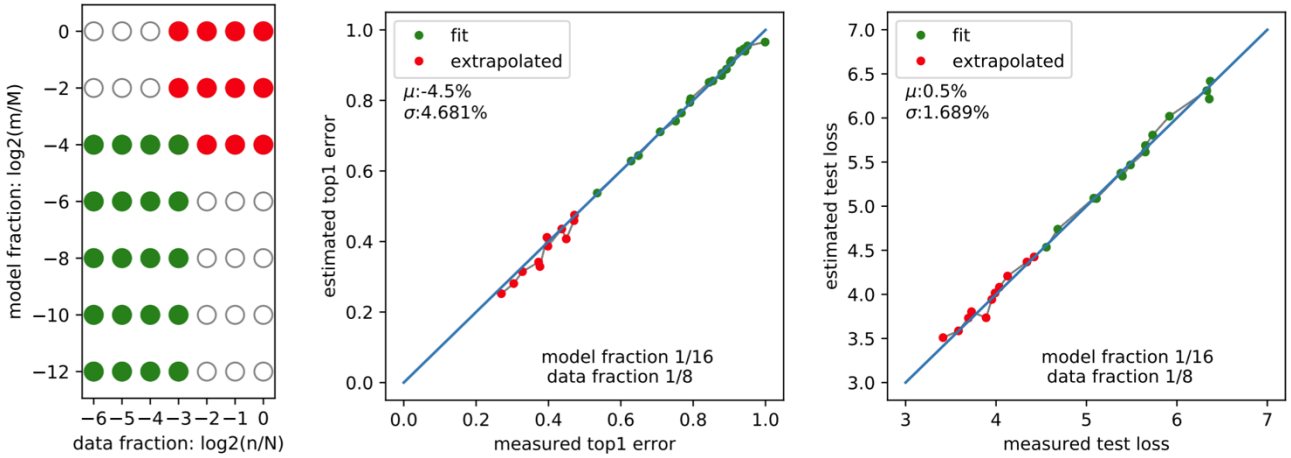
Provides surprisingly good fits to model-data joint error.



(a) Wiki103 error (cross entropy) landscape.

Model-data joint scaling is accurate

From Rosenfeld – fit scaling exponents on small data, small models. Predict rest.



(a) Illustration. (b) Extrapolation on ImageNet (c) Extrapolation on WikiText-103.

Trading off data size and model size: optimize $n^{-\alpha}+m^{-\beta}+C$ with your costs.

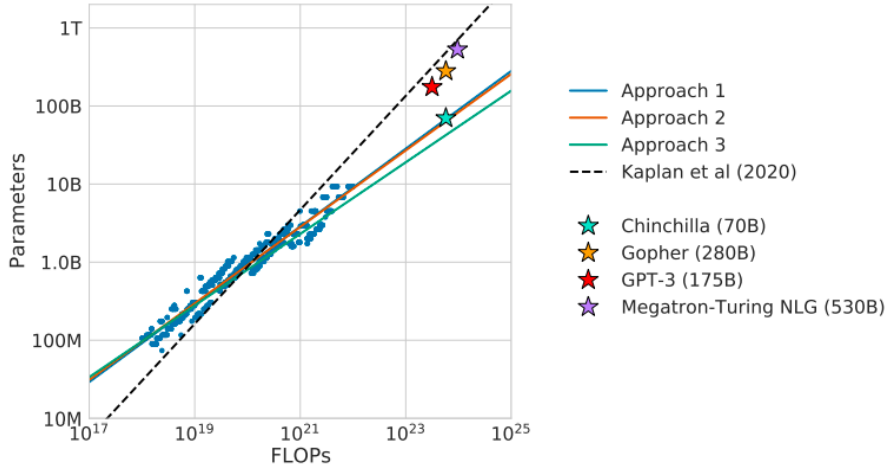
‘Optimal’ compute and data tradeoffs as a case study.

Rosenfeld, Kaplan both predict relationship of data, model and perf.

Kaplan claims: $N_{opt} = C^{0.73}, D_{opt} = C^{0.27}$ (tokens per param decreases w/ C)

N：模型参数量
D：训练数据量

Chinchilla [Hoffman et al] argue these fits are quite off.



Hoffman+ 2022

Chinchilla in depth – 3 methods

Approach	Coeff. a where $N_{opt} \propto C^a$	Coeff. b where $D_{opt} \propto C^b$
1. Minimum over training curves	0.50 (0.488, 0.502)	0.50 (0.501, 0.512)
2. IsoFLOP profiles	0.49 (0.462, 0.534)	0.51 (0.483, 0.529)
3. Parametric modelling of the loss	0.46 (0.454, 0.455)	0.54 (0.542, 0.543)
Kaplan et al. (2020)	0.73	0.27

The chinchilla authors suggest 3 ways of fitting scaling laws – we'll go over each.

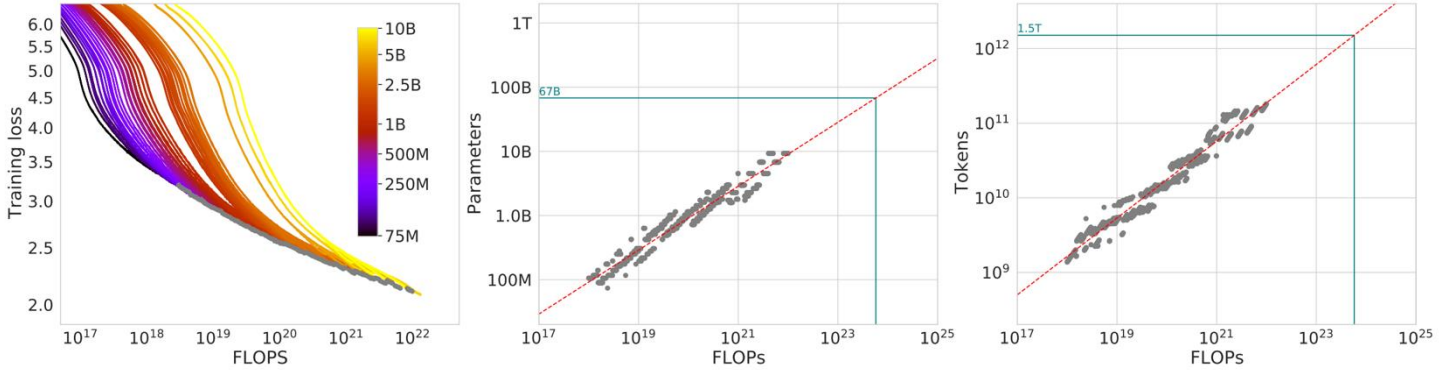
They mostly (minus method 3) suggest similar constants. More on this later..

Method 1 – minimum over runs.

就是上一页的approach 1

Similar to the FLOPS figure on Kaplan –
the minimum over the union of all training curves is a power law.

左图最下方的前沿，代表了给定flops，能达到的最小loss。对应中图的拟合曲线。



左图上标注出的灰点，就是中图中的那些点。

Figure 2 | **Training curve envelope.** On the **left** we show all of our different runs. We launched a range of model sizes going from 70M to 10B, each for four different cosine cycle lengths. From these curves, we extracted the envelope of minimal loss per FLOP, and we used these points to estimate the optimal model size (**center**) for a given compute budget and the optimal number of training tokens (**right**). In green, we show projections of optimal model size and training token count based on the number of FLOPs used to train *Gopher* (5.76×10^{23}).

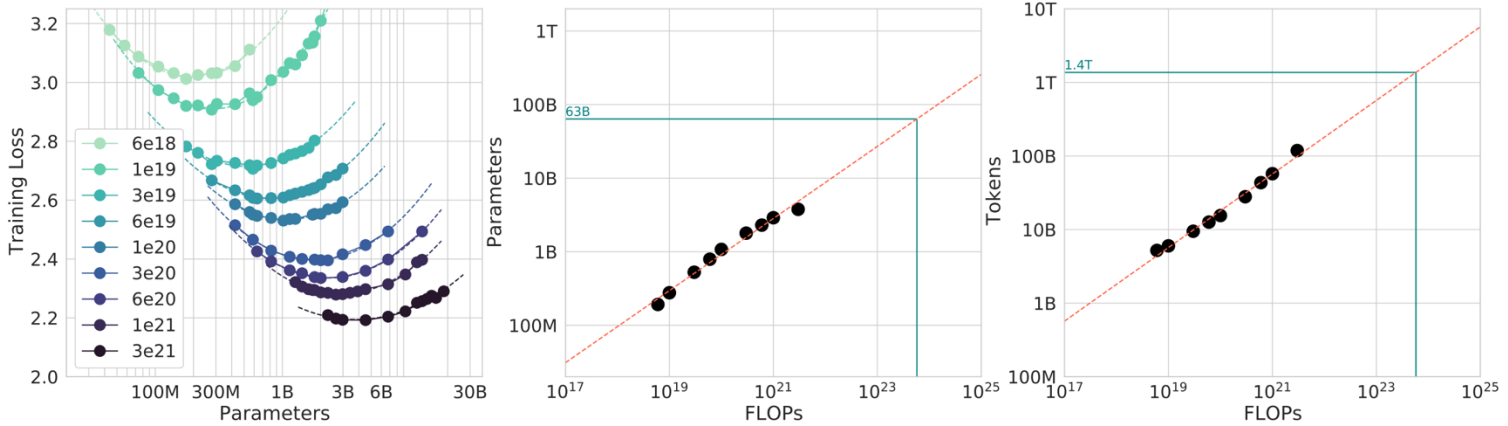
中图和右图是可以互相推导的。因为给定C，parameters增长，tokens就会减小，反之亦然。

Method 2 - IsoFLOPS

currently most popular method

通过中图和右图，我们可以预测，对于我们实际拥有的10^24算力，最佳的parameter是63B，最佳的数据量是1.4T token。

Pick a range of FLOP budgets, vary the total parameter count, take the min over these convex shapes. The minima form a power law.



先选定不同的flop budget（不同frontier curve），然后每个budget都做parameter-data trade-off，从而画出一条frontier

Figure 3 | **IsoFLOP curves.** For various model sizes, we choose the number of training tokens such that the final FLOPs is a constant. The cosine cycle length is set to match the target FLOP count. We find a clear valley in loss, meaning that for a given FLOP budget there is an optimal model to train (**left**). Using the location of these valleys, we project optimal model size and number of tokens for larger models (**center** and **right**). In green, we show the estimated number of parameters and tokens for an *optimal* model trained with the compute budget of *Gopher*.

Method 3 – Joint fits

Run a bunch of models on the size-data grid. Use least squares to fit a joint scaling law

拟合模型损失、data size、model size之间的关系，拟合出来后，就可以画出这里的两幅图。这里的两幅图本质上就是一个数据分布的两种可视化

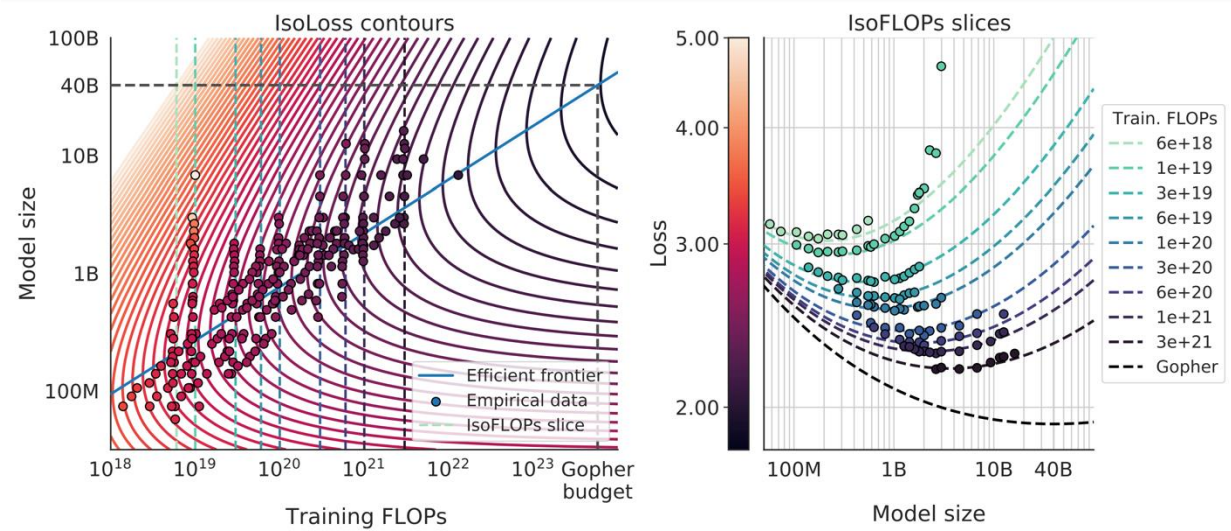
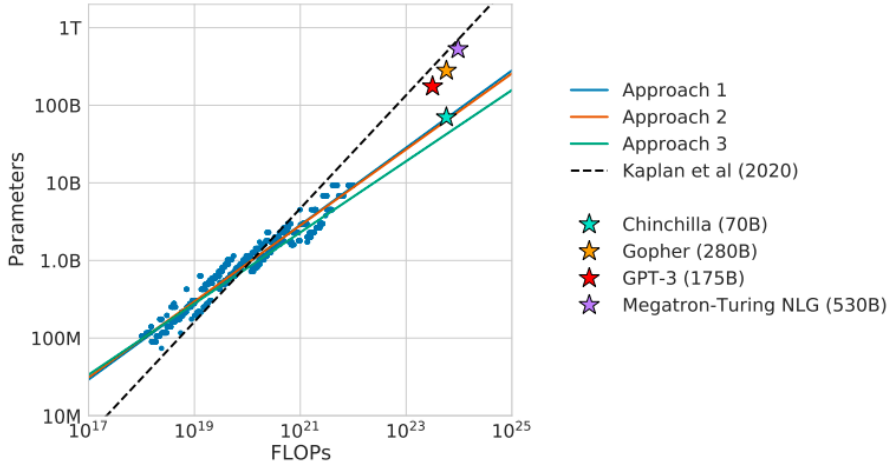


Figure 4 | **Parametric fit.** We fit a parametric modelling of the loss $\hat{L}(N, D)$ and display contour (**left**) and isoFLOP slices (**right**). For each isoFLOP slice, we include a corresponding dashed line in the left plot. In the left plot, we show the efficient frontier in blue, which is a line in log-log space. Specifically, the curve goes through each iso-loss contour at the point with the fewest FLOPs. We project the optimal model size given the *Gopher* FLOP budget to be 40B parameters.

Why do we have this big difference?

Kaplan claims: $N_{opt} = C^{0.73}, D_{opt} = C^{0.27}$ (tokens per param decreases w/ C)



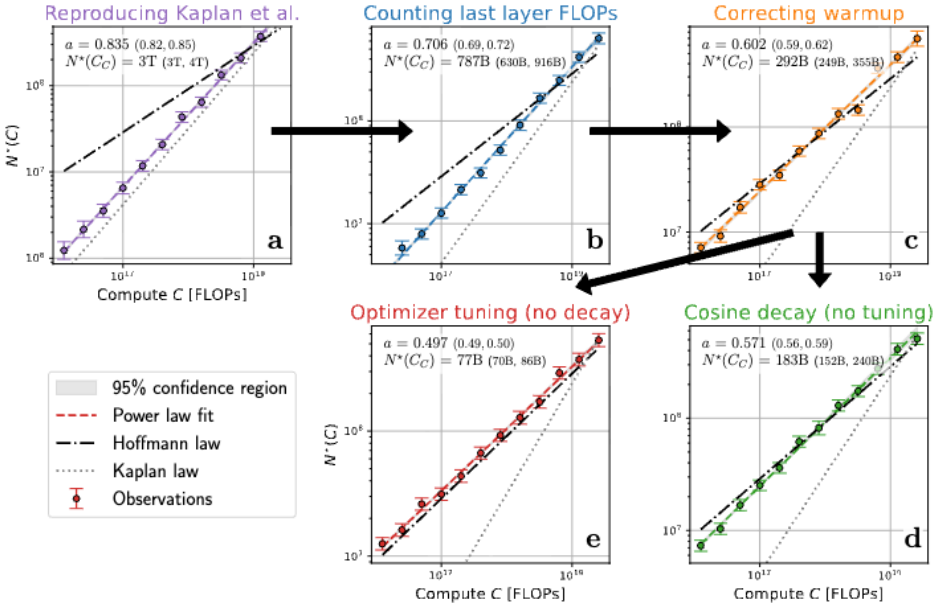
Why such a big difference (when both fit joint scaling laws?)

#TODO：这幅图中的曲线指的是在给定的横轴flops预算下，最佳的模型参数量吗？也就是让loss最小的模型参数量？
这里的不同指的具体是什么？kaplan和Chinchill两个研究产生的不同的scaling law吗？为什么会产生？结合本页和后面几页，详细解释下这块到底是在讲什么？

Explanation 1

Resolving Discrepancies in Compute-Optimal Scaling of Language Models

ll Wortsman† Jenia Jitsev‡ Ludwig Schmidt† Yair Carmon*



太多因素会影响到最后的scaling recipe了

- Kaplan removed last layer param from the count
- Warmup at very small compute budgets was too high
- (Decay itself is maybe not critical if batch / LR is properly tuned)

Explanation 2

Reconciling Kaplan and Chinchilla Scaling Laws

Tim Pearce *Microsoft Research*

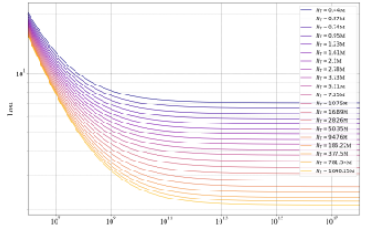
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Reviewed on OpenReview: *https://openreview.net/forum?id=NLoaLyuUUF*

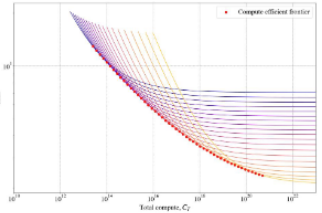
Start from fitted model of **Chinchilla's** training curves

$$\text{Loss}(N_T, D) = \frac{482}{N_T^{0.35}} + \frac{2085}{D^{0.37}} + 1.82$$

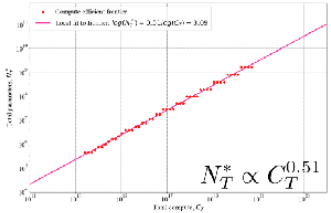
Generate training curves for model sizes used in **Kaplan's** study (1k to 1.5B params)



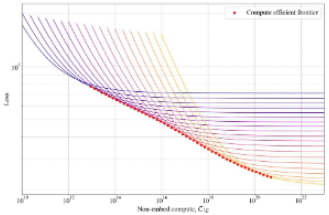
Find compute optimal frontier in terms of **total** parameters N_T and compute as in **Chinchilla**



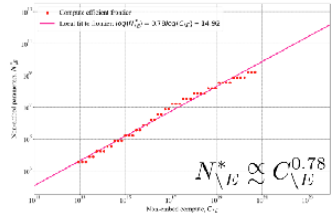
Find power law scaling coefficient of 0.51, close to Chinchilla's 0.50

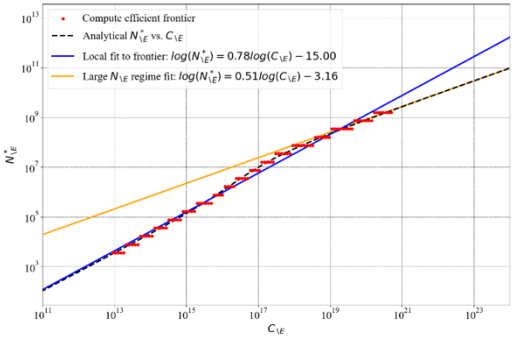


Find compute optimal frontier in **non-embedding** parameters $N_{\setminus E}$ and compute as in **Kaplan**



Find **local** power law scaling coefficient of 0.78, close to Kaplan's 0.73

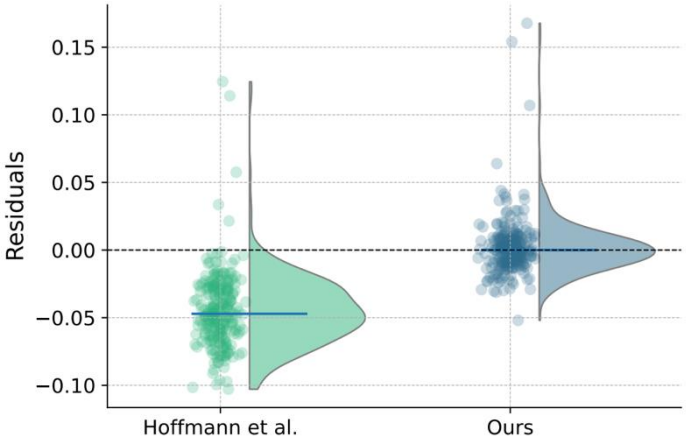


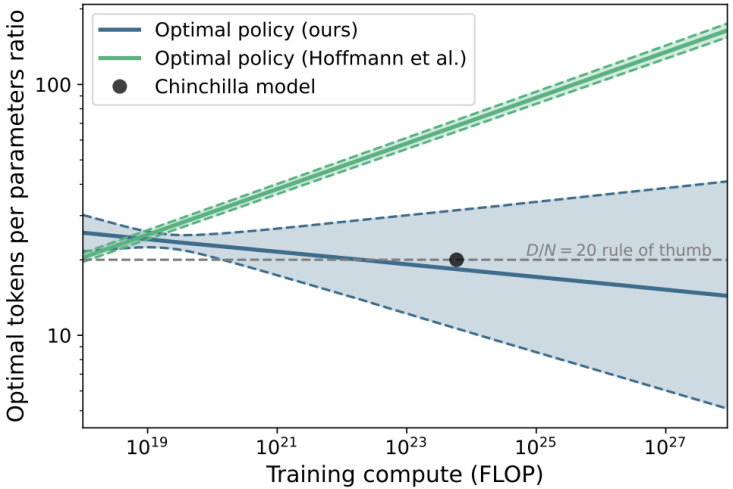


Non-embedding vs total param choice + small nonlinearities

Fun addendum – errors in chinchilla method 3

Note that this method three was likely flawed in the original paper. Some authors did data forensics, recovered the raw data, and re-did the fit and got results more consistent with methods 1 and 2





[Besiroglu et al 2024]

Important note – train-optimal is likely not what you want

Chinchilla aims to tell you what gives the best model for fixed training compute..

But most of the compute in a real deployment is inference.. So we should ‘over’ train

#TODO：这里为啥说要over train？这是啥意思，而且为啥要这样做?

- **GPT3** – 2 tokens / param
- **Chinchilla** – 20 tokens / param
- **LLaMA65B** – 22 tokens / param
- **Llama 2 70B** – 29 tokens / param
- **Mistral 7B** – 110 tokens / param
- **Llama 3 70B** – 215 tokens / param

… etc etc

The more usage we expect, the more it becomes worth it to pay the upfront cost

Isoflops everywhere

Methods like IsoFLOPS are pretty easy to execute and usually pretty clean

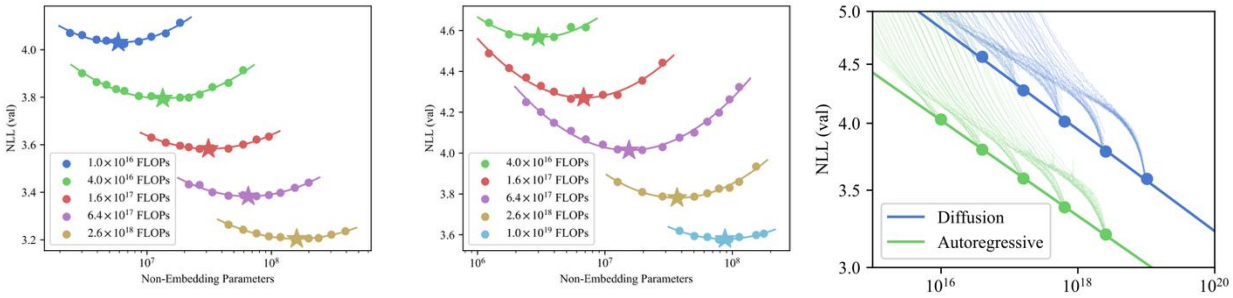
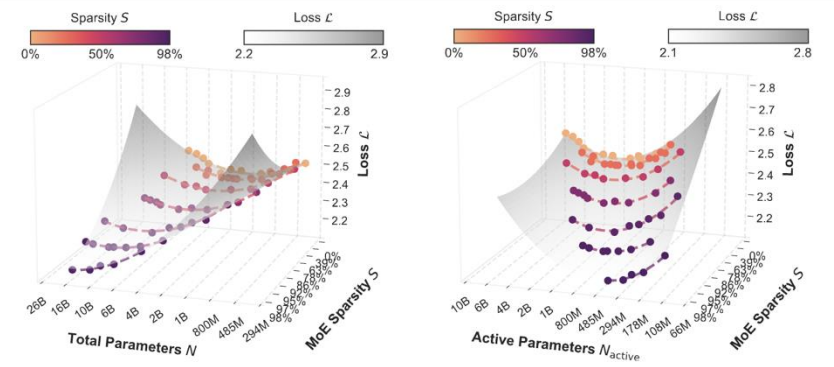


Figure 5: IsoFLOP profiles for autoregressive models (left) and diffusion models (right).

Diffusion - Gulrajani+ 2023.



(a) IsoFLOP surface over sparsity and total parameters (b) IsoFLOP surface over sparsity and active parameters

MoEs – Abnar+ 2025.

Scaling laws for models and compute

Log-linearity extends to model parameters and compute!

Lets us set the following based on small models

- Pick optimizer
- Pick architecture and model sizes

Also lets us make smart resource tradeoffs

- Big models vs more data?

Recap: scaling laws – surprising and useful!

- **Data scaling**: understand how data affects models, clean theory
- **Model scaling**: dramatically reduce costs for training
- **Scaling as prediction:** understand what problems can be ‘brute forced’

#TODO：为什么说model scaling可以降低训练成本?